

AI Semester Project Proposal

ReviewLens: Sentiment Analysis of Daraz Product Reviews Using Machine Learning



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1. Project Title

ReviewLens: An AI-Based Sentiment Analysis System for Daraz Product Reviews is a full-stack intelligent mobile application that combines **Artificial Intelligence (AI)** and **Mobile Application Development (MAD)** to analyze product reviews from Daraz and present sentiment insights through an interactive mobile interface.

The project consists of three major components:

2. Introduction

Online shopping platforms such as Daraz contain hundreds of customer reviews for each product. These reviews help buyers understand product quality, delivery performance, and seller reliability. However, manually reading all reviews is time-consuming and inefficient.

This project aims to build an Artificial Intelligence system that automatically analyzes product reviews and classifies them into three sentiment categories:

- Positive
- Negative
- Neutral

The system will scrape live reviews from Daraz, process the text using Natural Language Processing (NLP), and apply a Machine Learning model to predict sentiment. The final output will present an easy-to-understand summary showing sentiment percentages, common praises, and common complaints.

This project aims to develop an intelligent mobile application that automatically analyzes customer reviews from Daraz using Artificial Intelligence techniques and presents the results in a modern mobile interface.

The Artificial Intelligence component uses Machine Learning and Natural Language Processing to classify customer reviews into Positive, Negative, and Neutral categories. The backend system

scrapes live reviews from Daraz, processes them using a trained sentiment model, and returns structured insights such as sentiment percentages, common praise, and common complaints.

The Mobile Application Development component focuses on building a Flutter-based application that enables users to input a product URL and view the analysis results visually through charts, statistics, and review cards. By combining AI and mobile technologies, the project demonstrates a complete end-to-end intelligent application.

3. Problem Statement

Customers often face difficulty in evaluating products because:

- Products may contain hundreds of reviews.
- Reviews are written in both English and Roman Urdu.
- Some reviews are mixed or ambiguous.
- Important positive and negative opinions are hidden among many comments.

The proposed system will solve this problem by automatically extracting and summarizing customer sentiment.

4. Objectives

1. Scrape live product reviews from Daraz.
2. Preprocess review text using NLP techniques.
3. Train a Machine Learning model for sentiment classification.
4. Predict Positive, Negative, and Neutral sentiments.
5. Generate confidence scores for each prediction.
6. Build a FastAPI-based REST API.
7. Develop a Flutter mobile application.

8. Connect the Flutter app to the backend API.
9. Visualize sentiment summaries and charts in the mobile app.
10. Provide a practical solution for both AI and MAD coursework.

5. Dataset

The model will be trained using the **Amazon Product Reviews Dataset** available on [Kaggle](#).

Dataset Information

- Total Reviews: 568,000+
- Input Column: Text
- Rating Column: Score
- Labels:
 - 4–5 stars → Positive
 - 3 stars → Neutral
 - 1–2 stars → Negative

Although the dataset is from Amazon, it closely resembles e-commerce reviews found on Daraz and is suitable for training a general sentiment model.

6. Artificial Intelligence Technique

This project uses **Machine Learning for Text Classification**.

Model Used

- LinearSVC

Supporting Components

- CalibratedClassifierCV for confidence scores
- TfidfVectorizer for converting text into numerical features

Why Linear SVC?

Linear SVC is highly effective for text classification because it:

- Handles large sparse datasets efficiently
- Performs well on high-dimensional TF-IDF features
- Provides strong accuracy for sentiment analysis
- Works better than Logistic Regression on large review datasets

7. Natural Language Processing (NLP)

The following preprocessing steps will be applied:

1. Convert text to lowercase
2. Remove URLs and HTML tags
3. Remove special characters
4. Normalize spaces
5. Extract TF-IDF features

In addition, a hybrid rule-based layer will use:

- Product star ratings
- Roman Urdu sentiment keywords

to improve predictions on real Daraz reviews.

8. Methodology

Step 1: Data Preparation

- Load and clean the Amazon review dataset
- Convert ratings into sentiment labels

Step 2: Class Balancing

- Undersample positive reviews
- Oversample neutral reviews

Step 3: Model Training

- Train Linear SVC using TF-IDF features
- Calibrate the model to produce confidence probabilities

Step 4: Evaluation

- Measure Accuracy, Precision, Recall, and F1-score

Step 5: Artificial Intelligence Module

- Collects and preprocesses customer reviews.
- Uses Natural Language Processing (NLP) and Machine Learning.
- Classifies reviews as Positive, Negative, or Neutral.
- Generates confidence scores and keyword-based summaries.

Step 6: Application Module

- Developed using [Flutter](#) and [Dart](#).
- Allows users to enter a Daraz product URL.
- Sends requests to the FastAPI backend.
- Displays sentiment charts, review summaries, and confidence scores.
- Provides a user-friendly interface for mobile devices.

Step 7: Deployment

- Save the trained model as sentiment_model.pkl

Step 8: Live Analysis

- Scrape Daraz reviews and classify them using the trained model

9. System Architecture

1. User provides a Daraz product URL.
2. Python scraper extracts customer reviews.
3. Reviews are cleaned and transformed.
4. The machine learning model predicts sentiment.
5. Rule-based corrections are applied.
6. Results are returned as JSON.
7. A mobile application displays charts and summaries.

10. Technologies and Tools

Artificial Intelligence

- [Python](#)
- [pandas](#)
- [scikit-learn](#)
- [Kaggle Notebooks](#)

Backend Development

- [FastAPI](#)
- [joblib](#)
- [cloudscraper](#)

Mobile Application Development

- [Flutter](#)
- [Dart](#)
- [HTTP package for Flutter](#)
- [FL Chart](#)

11. Expected Output

The system will provide:

- Total number of analyzed reviews
- Percentage of positive, negative, and neutral reviews
- Common praise keywords
- Common complaint keywords
- Sample reviews with confidence scores

12. Expected Results

The trained model is expected to achieve approximately:

- Accuracy: 78%–85%
- Improved neutral sentiment detection
- Better performance on live Daraz reviews using hybrid correction rules

13. Applications

This system can be used by:

- Online shoppers
- E-commerce analysts
- Product managers
- Sellers and vendors
- Market researchers

14. Future Enhancements

- Support for Urdu text
- Transformer models such as BERT
- Cloud deployment
- Database integration
- Comparison of multiple products

15. Conclusion

ReviewLens is an end-to-end Artificial Intelligence project that combines Machine Learning, Natural Language Processing, web scraping, and API development to automatically analyze e-commerce reviews. The project demonstrates how AI can transform large volumes of unstructured customer feedback into meaningful insights that help users make better purchasing decisions.